Week 1 Documentation
Data Preparation & Metadata

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Contents

1 Objective

The goal of this task is to:

- Load legal datasets using streaming
- Sample documents efficiently
- Create structured metadata
- Save metadata in CSV format
- Organize datasets for the RAG pipeline

2 Datasets Used

The following datasets were used:

- Multi Legal Pile
- Pile of Law
- EUR-Lex
- policy QA

These datasets contain legal text that will be used later for retrieval and question answering.

3 Environment Setup

3.1 Python Version

```
Python 3.11
```

3.2 Main Libraries

```
datasets
pandas
itertools
```

3.3 Virtual Environment

```
HR_RAG (conda environment)
```

4 Data Loading Strategy

Because the datasets are very large, streaming mode was used:

```
dataset = load_dataset(
    "...",
    split="train",
    streaming=True
)
```

4.1 Why Streaming?

- Avoids full download
- Reduces RAM usage
- Faster experimentation
- Suitable for large-scale RAG pipelines

5 Sampling Strategy

A subset of documents was selected using:

```
from itertools import islice
sample_docs = list(islice(dataset_stream, N))
```

5.1 Sample Size Decision

The value of N was chosen based on:

- Available RAM
- Processing speed
- Need for quick experimentation

Chosen value:

```
N = 1000
```

5.2 Justification

The selected sample size provides a balance between computational efficiency and dataset coverage, enabling fast experimentation without exhausting system resources.

6 Metadata Design

Each document is converted into a metadata dictionary.

6.1 Metadata Fields

Field	Description
id	Unique document identifier
source	Dataset name
text_length	Length of document
dataset_split	Train/test split
language	Document language

6.2 Example Metadata Structure

```
metadata = {
    "id": idx,
    "source": "pile_of_law",
    "text_length": len(text),
    "dataset_split": "train",
    "language": "en"
}
```

7 Metadata Storage (CSV)

Metadata from all datasets was combined into a single CSV file.

7.1 Folder Structure

```
data/
     row
              pile_of_law/
              eur/
      metadata      .csv
```

7.2 Saving Code

```
import pandas as pd

df = pd.DataFrame(metadata_list)
df.to_csv("metadata.csv", index=False)
```

8 Challenges Faced

8.1 Issue 1 — DLL Error with PyTorch

Problem: WinError 1114 when loading torch.
Solution:

- Installed compatible CUDA/CPU version
- Verified environment configuration

8.2 Issue 2 — DatasetNotFoundError

Problem: Some datasets were unavailable.
Solution:

- Verified dataset names
- Switched to accessible datasets
- Used streaming mode

8.3 Issue 3 — ReadTimeout

Problem: Timeout while streaming large files.
Solution:

- Reduced sample size
- Retried download
- Used smaller batches

9 Next Steps

In Week 2, the pipeline will:

- Clean and chunk documents
- Generate embeddings
- Build vector database
- Implement retrieval